

$$I_1 = 9375 \text{ N}$$

4 (a) (i) Diffraction refers to the spreading of light waves when they pass through each of the apertures of the diffraction grating. Diffraction is significant and observable if the width of the apertures is of the same order of magnitude as the wavelength of the wave.

(ii) Two light waves emerging from the apertures of the diffraction grating have coherence if they have a constant phase difference.

**Same phase difference is not an acceptable answer.*

(iii) The principle of superposition states that when two or more waves of the same kind, such as those light waves emerging from the apertures, meet at a point in space, the resultant displacement of the waves at that point is the vector sum of the displacements of the individual waves.

**A common mistake made is many stated that the resultant wave had an amplitude that was the sum of the amplitudes of the overlapping waves.*

**For the whole of part (a), a common mistake made is that the question was not answered related to the terms given in the context of the question.*

(b) (i) $d \sin \theta = n\lambda$

$$\left(\frac{1}{500000}\right) \sin \theta = 2(650 \times 10^{-9})$$

$$\theta = 40.5^\circ$$

(ii) $d \sin \theta = n\lambda$

$$\left(\frac{1}{500000}\right) \sin \theta = 3(350 \times 10^{-9})$$

$$\theta = 31.7^\circ$$

(iii) There is an angular overlap of 8.8° between the second and third order maxima. Within this 8.8° , the colours are mixed up and it is difficult to tell which colour belongs to which order of the spectrum.

5 (a) $F = \frac{Q_1 Q_2}{r^2}$